

Enterprise Decision Intelligence Framework

Version 1.0

June 2026

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Abstract

Organizations today generate vast amounts of operational, workforce, financial, and customer data. However, many enterprises continue to struggle with transforming data into actionable decisions. Enterprise Decision Intelligence (EDI) provides a structured approach that combines data intelligence, knowledge systems, advanced analytics, and artificial intelligence to support strategic, operational, and workforce-related decision-making.

This paper presents a conceptual Enterprise Decision Intelligence Framework that integrates enterprise data sources, AI-driven analytics, knowledge management systems, and Agentic AI capabilities to improve organizational effectiveness and business outcomes.

1. Introduction

Modern enterprises operate in increasingly complex environments characterized by rapid technological change, workforce challenges, market uncertainty, and growing data volumes.

Traditional reporting systems often provide visibility into historical events but lack the intelligence required to support proactive and predictive decision-making.

Enterprise Decision Intelligence addresses this gap by combining:

- Enterprise Data Management
- Knowledge Intelligence
- Predictive Analytics
- Prescriptive Analytics
- Generative AI
- Human-in-the-Loop Governance

The objective is to enable organizations to make faster, more informed, and context-aware decisions.

2. Enterprise Decision Intelligence Framework

The proposed framework consists of five interconnected layers:

2.1 Enterprise Data Sources

The framework integrates data from multiple enterprise systems, including:

- Human Resources
- Finance
- Customer Relationship Management (CRM)
- Enterprise Resource Planning (ERP)
- Project Management Offices (PMO)
- Operational Systems

These systems provide structured and unstructured data required for decision-making.

2.2 Data Intelligence Layer

The Data Intelligence Layer performs:

- Data Integration
- Data Cleansing
- Deduplication
- Aggregation
- Transformation

This layer establishes a trusted enterprise data foundation.

2.3 Knowledge Intelligence Layer

The Knowledge Intelligence Layer transforms enterprise data into organizational knowledge through:

- Semantic Models
- Knowledge Graphs
- Contextual Relationships
- Enterprise Knowledge Repositories

This enables contextual understanding beyond raw data analysis.

2.4 Decision Intelligence Engine

The Decision Intelligence Engine combines multiple analytical approaches:

Predictive Analytics

Forecasts future outcomes and identifies potential risks.

Prescriptive Analytics

Recommends optimized actions and decision paths.

Generative AI Insights

Produces contextual recommendations, summaries, explanations, and decision support narratives.

2.5 Enterprise Decision Hub

The Enterprise Decision Hub acts as the central coordination point where insights are transformed into actionable business decisions.

Key decision domains include:

- Workforce Decisions
- Operational Decisions
- Strategic Decisions

3. Agentic AI Extension

To further enhance decision-making, the framework incorporates Agentic AI.

Specialized AI agents perform domain-specific functions:

- Resource Agent
- Talent Agent
- Knowledge Agent
- Operations Agent

These agents collaborate through an AI Orchestrator and Decision Agent Layer.

Human decision-makers remain responsible for final approval and governance, ensuring transparency and accountability.

4. Business Applications

Potential enterprise applications include:

Workforce Optimization

- Talent Planning
- Skills Intelligence
- Resource Allocation

Enterprise Operations

- Service Delivery Management
- Process Optimization
- Operational Risk Management

Strategic Planning

- Investment Decisions
- Growth Strategies
- Market Expansion

5. Future Research Directions

Future research areas include:

- Multi-Agent Enterprise Ecosystems
- Enterprise Digital Twins
- AI Governance Frameworks
- Autonomous Decision Support Systems
- Human-AI Collaborative Intelligence

Conclusion

Enterprise Decision Intelligence represents the next evolution of enterprise decision-making. By integrating enterprise data, knowledge systems, analytics, and Agentic AI capabilities, organizations can move beyond traditional reporting toward intelligent, proactive, and adaptive decision support systems.

The proposed framework provides a foundation for future research and practical implementation of AI-driven enterprise decision ecosystems.